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Research paper

A deep learning based relative clarity classification method for infrared and visible image fusion

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ABSTRACT

Infrared and visible image fused(IVIF) results normally suffer from detail loss, noise occurrence, low contrast, and blurred edges. In this paper, a new method is proposed to address the detail loss, low contrast, and blurring issue in IVIF. Specifically, visible images are enhanced by guided filter and high dynamic range compression. Infrared images are normalized by a linear transformation. Then we use blur and clear discrimination to detect salient pixels between infrared and visible images. A fully weight shared multi-path residual neural network is proposed for blur discrimination between infrared and visible image pixels in the same position. Clear pixels are treated as salient pixels, which contribute more to fused images than blur pixels. The output of our proposed network is a binary classification map for blur and clear discrimination, which is treated as our fusion weight map in the fusion stage. To deal with the discontinuous problem, we compute the two distance-transformed maps of the binary classification map and its complementary map. The two distance-transformed maps are used as weight maps to fuse the enhanced infrared and visible images. Finally, we use single scale retinex (SSR) to further enhance our fused images. The experimental results in public IVIF datasets demonstrate the superior performance of our proposed approach over other state-of-the-art methods in terms of both subjective visual quality and objective metrics. The source code is available in https://github.com/eyob12/Multi_path_residual_neural_network_based_IVIF

1. Introduction

Normally, the goal of infrared and visible image fusion (IVIF) is to construct a fused image that preserves both the saliency of targets in the infrared image and the abundant detail information in the visible image. It has wide applications in remote sensing, medical diagnosis, surveillance, military, machine vision, and photography.

Generally, the IVIF method includes fusion feature extraction, feature fusion, and fusion image generation. A very simple IVIF method is taking the pixel-wise maximum between two images, where feature extraction is the identify transformation, feature fusion is the maximization operation, and fusion image generation is also the identify transformation. Different IVIF methods vary among the three stages.

There exist many IVIF methods [1,2]. They can be classified to two classes: hand designed feature based methods and feature learning based approaches. The traditional techniques include multi-scale transformation, sparse representation, subspace and saliency methods [1,2], Discrete Wavelet Transformation (DWT) [3], Discrete Cosine Harmonic Wavelet Transform (DCHWT) [4], Gradient Transfer and Total Variation Minimization (GTF) [5,6], Adaptive Sparse Representation (ASR) [7], low rank decomposition [8], Non-subsampled Contour Transformation (NSCT) [9], and Nonsubsampled Pyramid Filter (NSP) [10]. The modern approaches include deep learning based methods in [11–21]. Due to the unavailability of ground truth fusion images, most of deep learning based methods use self-supervised strategy, in which the fused image of deep network is constrained to look similar to both infrared and visible images. Therefore, the fused image cannot be guaranteed to contain all salient objects from both infrared and visible images.

In this paper, a pair of two ordered pixels (for example, visible pixel in the first and infrared pixel in the second) in the same position from visible and infrared images is classified into two classes by deep learning (1 for clearer visible pixel and 0 for clearer infrared pixel). The clearer pixel should contribute more to fused image than less clear pixel. A fully weight shared multi-path residual neural network is proposed for the binary classification of each pair of pixels. To deal with possible boundary discontinuous problem with binary classification, we compute distance transformed maps for both the binary classification

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Fig. 1. Schematic illustration of image fusion. From left to right: (A) infrared image, (B) visible image, (C) DTF-SR [22], (D) RFN-Nest [21], (E) FusionGAN [17], and (F) Our method.

map and its complementary map. Then we enhance original infrared and visible images. The two distance transformed maps are used as weight map to fuse the enhanced infrared and visible images. Finally, we use single scale retinex (SSR) to further enhance our fused images. We prepare our training data by simulating blurry versions of clear images from another field.

To demonstrate the significant superiority of our approach, we present a representative example in Fig. 1. The first two images illustrate the infrared and visible images to be fused. The visible image showcases a detailed background, including a house, table, and standing chairs, while the infrared image highlights the targets, namely persons. The third image shows the fusion outcome of a recent method, DTF-SR [22]. This conventional method preserves thermal radiation from the infrared images but sacrifices texture information from the visible image, as indicated in the red box (i.e., table and stand chair).

In comparison, the fourth image displays the fusion outcome by the RFN-Nest method [21]. This method lost the thermal radiation information from the infrared image, i.e., persons in the upper-right corner and blurred targets. Furthermore, it also loses some texture information from the visible image, as shown in the red box (i.e., table and stand chair). Similarly, the fifth image shows the fusion outcome by FusionGAN [17], which also suffers from the loss of thermal radiation information from the infrared image, i.e., persons in the upper-right corner and blurred targets. Additionally, it totally loses texture information from the visible image, as indicated in the red box (i.e., table and stand chair).

In contrast, the rightmost image in Fig. 1 displays the fusion outcome using our method. Our approach effectively preserves the thermal radiation distribution in the infrared image, facilitating easy target detection. Simultaneously, it retains the details of the background (i.e., house, road, table, and standing chairs) in the visible image, showcasing its comprehensive preservation capabilities. The main contributions of our work in this paper are:

- converting IR and VIS image fusion problem into a clear and blur discrimination problem
- proposing a new fully automatic deep learning and image enhancement based IVIF system
- designing a fully weight shared multi-path residual neural network for clear and blur classification.
- verifying the superior performance of our proposed method in public datasets and providing source codes for research community.

The rest of this paper is structured as follows. In Section 2, Related work and motivation is given. Our proposed method is in Section 3. Experimental results and discussion are shown in Section 4. Finally, we draw some conclusion in Section 5.

2. Related work and motivation

Infrared and visible image fusion techniques have become a popular research field in various types of applications, such as remote sensing, medical diagnosis, surveillance, military, machine vision, and photography applications. A review and application review of recent IR and VIS image fusion techniques is summarized in [1,2]. The goal of infrared and visible image fusion (IVIF) is to create an image that combines the saliency of targets in the infrared image with the abundant detail information in the visible image. In the last decade, many methods have been proposed to fuse infrared and visible images. But generally, we can categorize fusion algorithms into seven categories based on their theories.

(1) Multi-scale transform: Over the past decade, the multi-scale transform method has been utilized in infrared and visible image fusion. This process begins by decomposing the original image into various components and then fusing the multi-scale representation of the source image according to a given fusion rule. Then the fused result is obtained through inverse multi-scale transform. Multi-scale transforms such as Discrete Wavelet Transform (DWT) [3], Discrete Cosine Harmonic Wavelet Transform (DCHWT) [4], Nonsubsampled Contour Transformation(NSCT) [9], and Nonsubsampled Pyramid filter (NSP) [10] are commonly employed in image fusion.

(2) Sparse representation [23–25]: The process of sparse representation starts with breaking down the source image into multiple overlapping patches. Then, a sliding window approach, an over-complete dictionary, and sparse coding are utilized to construct the coefficient of the sparse representation. These coefficients are then merged using a pre-determined fusion law. Lastly, the fused coefficients and the over-complete dictionary are used to reconstruct the fused image.

(3) The subspace-based method [26,27]: Subspace-based methods, such as PCA, ICA, and NMF, have proven successful in both infrared and visible image fusion. These methods involve projecting high-dimensional input images onto low-dimensional spaces or subspaces. This technique yields effective results and is widely utilized in the field.

(4) Salience-based fusion method [28,29]: The saliency fusion technique also employs a multi-scale transform to separate the original image into a base layer and a detail layer, after that saliency extraction models are applied to the base or/and detail layer of the source image to generate the saliency map. Then, the weight map can be obtained, and finally, the fused image can be subsequently constructed.

(5) Neural network based [30,31]: The neural network usually consists of many neurons and can thus imitate the perception behavior mechanism of the human brain to deal with neuron information. In recent years, the deep learning approach has demonstrated state-of-the-art performance in infrared and visible image fusion.

(6) The hybrid fusion method [32,33]: This hybrid method integrates other types of techniques into the multi-scale transform framework, such as hybrid multi-scale transform and saliency, hybrid multi-scale transforms and sparse representation, hybrid multi-scale transforms and neural network, hybrid Salience Based Fusion and neural network.

(7) Other fusion methods [6,34]: Other infrared and visible image fusion methods can inspire new ideas and perspectives for image fusion. Some of the other types of infrared and visible image fusion methods are entropy, Markov random field, morphology, and infrared feature extraction, and visual information preservation.

Over the past decade, numerous methods for fusing infrared and visible images have been developed. These methods can be broadly categorized into two classes based on their feature extraction techniques: traditional non-feature-learning-based methods and modern feature-learning-based approaches. The traditional non-deep-feature-learning fusion techniques such as multi-scale transformation, sparse representation, subspace, and saliency methods. The modern "deep feature learning fusion" approaches refer to deep neural network-based methods. The traditional fusion methods use hand crafted features in feature extraction stage, however, modern fusion approaches take advantage of the automatic feature learning power of deep convolutional neural network. Therefore, the modern fusion approaches based on deep networks have a large improvement in the feature extraction stage over traditional methods.

In deep learning, there are three commonly used networks for IVIF purposes, namely those based on the Convolutional Neural Network (CNN), the Autoencoder (AE), and the Generative Adversarial Network (GAN). Convolutional Neural Networks (CNNs) have played a leading role in image fusion, showcasing their significance through influential models such as IFCNN [35], MDLatLRR [36], and NestFuse [37]. Their strength lies in the ability to learn hierarchical feature representations, effectively capturing both spatial and spectral information. However, CNNs are susceptible to overfitting with limited data and may struggle when faced with diverse image characteristics. In image fusion, for the first time, Liu, Yu, et al. [11] proposed a convolutional neural network (CNN)-based fusion method. To train the network and generate a decision map, they used image patches containing different blur versions of the input image. However, this method is not well-suited for infrared and visible images as it was designed for multi-focus image fusion. In addition, Li, Hui [12] proposed a simple and effective infrared and visible image fusion method based on a deep learning framework. However, the fused images suffer losing of detailed information, fusion noise, low contrast effect and blurred edge of fused objects.

Similarly, Liu, Yu, et al. [14] proposed a convolutional neural network-based infrared and visible image fusion method. This method uses a Siamese network to obtain the network weight map. The weight map combines the pixel activity information from the two sources. The model has mainly divided into four steps: the infrared image and the visible image are passed into the convolutional neural network to generate weights; the Gaussian pyramid is used to decompose the weight of the source image, and the two source images are decomposed by the Laplacian pyramid, respectively. The information obtained by the decomposition of each pyramid is fused with the coefficients in a weighted average manner. The limitation is that it cannot be combined effectively with conventional fusion technology and is not suitable for complex datasets.

Generative Adversarial Networks (GANs) represent a more recent breakthrough, recognized for their ability to generate realistic images. This makes them particularly well-suited for tasks involving image fusion. DDcGAN [19] and MEF-GAN [38] are notable examples. One of their notable strengths is their capability to handle complex details and textures, which results in visually appealing fused images. However, it is important to note that generative adversarial network (GAN) training can be unstable, which can potentially introduce unwanted artifacts. Additionally, utilizing GANs for image fusion tasks can incur significant computational expenses. For example, FusionGan [17] was the first method to fuse infrared and visible images using GAN. By adding the source images to the GAN network generator, the generator produced a fused image that contains the source image feature information. It designs the appropriate loss function to control the structure of the fused image in the generator, and the discriminator improves the fusion quality. However, the fused image is unstable, and the fusion effect is not natural. Although the feature information in infrared images can be extracted, the fused image loses the edge and detail texture information of the source images.

Autoencoders are a class of neural networks that play a pivotal role in unsupervised learning and feature representation, which is designed to learn efficient representations of input data by encoding them into a lower-dimensional space and subsequently reconstructing the original input from this encoded representation. However, it is important to note that they may experience information loss during compression, and their capability to handle complex image structures is somewhat limited. In the autoencoder methods, the encoder is dedicated to extracting effective features from the input image, while the decoder reconstructs the input image from the encoded features. Then it is natural that the trained autoencoder can be used to solve two subproblems in image fusion: feature extraction and image reconstruction. Therefore, the key to image fusion lies in the design of feature fusion strategies. DenseFuse [39] is one of the most famous AE-based methods, which trains the encoder and decoder on the COCO dataset and adopts the addition and L1-norm fusion strategies. At present, in infrared and visible image fusion, the strategies for feature fusion are still hand-calculated and not learnable, such as addition, L1 norm, and attention weighting. This hand-calculated fusion strategy is rough, which limits the further improvement of infrared and visible image fusion.

Unsupervised learning methods, like U2Fusion [40], provide an efficient way to learn without labeled data. Although beneficial in situations with limited resources, it is worth noting that these methods might not reach the same level of performance as supervised approaches and have restricted control over specific fusion objectives.

Recently, Ma et al. [41] proposed a general image fusion framework based on cross-domain long-range learning and Swin Transformer, termed as SwinFusion, which achieved sufficient integration of complementary information and global interaction. It was demonstrated that the global attention mechanism was helpful in improving fusion performance in [41]. Rather than only focusing on one fusion task, Tang et al. [42] incorporated image registration, image fusion, and semantic requirements of high-level vision tasks into a single framework and proposed an image registration and fusion method, named SuperFusion, which considered multiple tasks simultaneously.

The methods such as [12–15], and [16] have been proposed for image fusion. These methods assume that informative pixels in a deep feature map have a higher response amplitude than uninformative pixels. They calculate a weight map based on the response amplitude of the feature map and use a weighted averaging strategy for fusion. Different from the weight map generation approach, some authors proposed generative adversarial network (GAN) based fusion methods [17–19,43] or auto-encoder based fusion approaches [20,21], where the fusion strategy is part of the network and can be trained. However, because of the lack of ground truth, input images are usually used as supervised information in most of deep learning based methods [11–21,44].

In [11,44], the multi-focus image fusion problem is solved by a blur classification method. Inspired by it, we convert the IR and VIS image fusion problem into a clear and blur discrimination problem and propose a new IVIF method combining deep learning and image enhancement. The methods for multi-focus image fusion [11,44] cannot be simply extended to solve IVIF problem because multi-focus image fusion methods solve focus problem and IVIF methods solve multi-modality (visible and infrared) image fusion problem. Our work in this paper is mainly different from [11,44] in the following: (1) We solve the infrared and visible image fusion problem rather than multi-focus image fusion problem; (2) We propose an IVIF system based on deep learning and image enhancement; (3) We propose a fully weight shared multi-path residual network for clear and blur pixel classification; (4) Experimentally, our proposed method is superior to the methods in [11,44] which are tailored to multi-focus image fusion problem.

3. Proposed method

Our proposed method mainly consists of five steps: (1) simulating and collecting training data; (2) designing and training a fully weight shared multi-path residual neural network for clear or blur classification; (3) enhancing infrared and visible input images; (4) fusing enhanced IR and VIS images ; and (5) finally enhancing fused images. The diagram of our proposed IVIF method is shown in Fig. 2. The left part in Fig. 2 shows the network training phase with input patch pairs. The two patches in each pair centered around the same position in

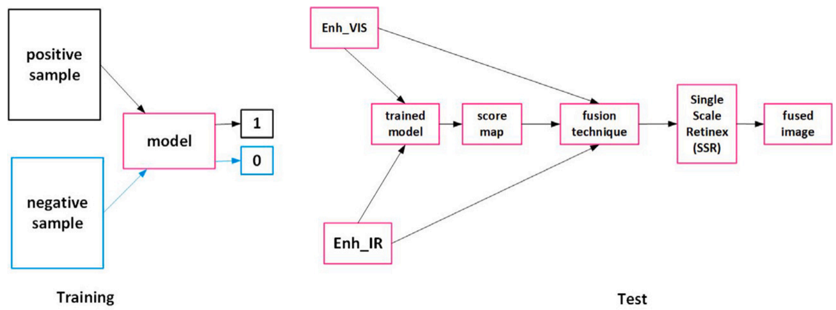

Fig. 2. Schematic diagram of our proposed IVIF method.

clear and blurry images, respectively. They are concatenated in row as an input. The pair is labeled as 1 if the upper patch is clearer than the lower patch. Otherwise, it is labeled as 0. The trained network is able to compare the relative clarity between any two patches. In the right part in Fig. 2, the trained network computes binary relative clarity map as fusion weighting map. The weighted average of the enhanced infrared and visible images is computed by using the weighting map from the neural network output. And the weighted average map is finally enhanced by single scale retinex (SSR) as the final fused image. More details are given below sections.

3.1. Proposed weight sharing multi-path neural network

In this paper, two patches centered around the same position from visible image and corresponding infrared image are denoted by V^0 and I^0, respectively. We also use the following symbols: V^h for horizontal derivative of V^0, V^v for vertical derivative of V^0, I^h for horizontal derivative of I^0, I^v for vertical derivative of I^0. We take the assumption that relatively clearer pixel means more informative pixel and it should contribute more to fused image. With this assumption, the IVIF problem is converted to a relative clarity comparison problem between V^0 and I^0. We solve the comparison problem by deep binary classification method. Specifically, the tuple $t = (V^0, I^0, V^h, I^h, V^v, I^v)$, consisting of six patches, is labeled as 1 if V^0 is clearer than I^0. Otherwise, it is labeled as 0. The tuple t is the input of our proposed deep neural network.

We propose a fully weight sharing multi-path residual convolutional network for relative clarity classification as shown in Fig. 3. As shown in Fig. 3(a), our overall network contains 6 paths for 6 patch inputs (size 32×32) V^0, I^0, V^h, I^h, V^v, and I^v, respectively. The outputs (size $4 \times 4 \times 256$) of the six paths are inputed to a concatenation layer(output size $4 \times 4 \times 1536$) followed by two fully connected layers FC_1(output size 120) and FC_2 (output size 1). The last layer is a sigmoid layer outputting a relative clarity score number. To reduce the number of filters, the six paths have the same structure as shown in Fig. 3(b): sequential organization of six layers: Convolutional layer, Max Pooling, Identical Residual Block(IDResB), Convolutional Residual Block (ConvResB), Identical Residual Block(IDResB), Convolutional Residual Block (ConvResB). The six paths does not only share structure but also share weights. It means that each of the six path is totally the same. One may think that the six paths can be implemented by a six time loop over one path. However, we organize the six paths parallelly for efficient computation by taking advantage of GPU parallel computation capability. The parallel organization also makes method implementation easy in the common modern deep learning library. We also propose IDResB and ConvResB as shown in Fig. 3(c) and (d), respectively. Our IDResB contains five layers (Convolution, Batch Normalization (BN), Convolution, Batch Normalization (BN), and ReLU) and has a identity residual structure. IDResB keep the same data dimension as input. Our ConvResB possesses six layers (Convolution, Batch Normalization (BN), Convolution, Batch Normalization (BN), ReLU, and Pooling) and has a down sampling residual structure (Consisting of a convolution layer and a pooling layer). IDResB reduces spatial dimension to half and doubles the channel dimension. The ideal output “1” of our proposed network means V^0 is relatively clearer than I^0. The ideal output “0” means V^0 is relatively less clear than I^0).

Our proposed network features multi-path, weight sharing and residual structure. Although the six paths are the same, we input six patches to six paths for efficient computation. It is also possible to concatenate all patches in the third dimension to use one path. However, our parallel organization and weight sharing structure in this paper has less number of filters.

3.2. Cross domain training strategy

Normally, IVIF datasets do not provide label information about the relative clarity between visible pixels and infrared pixels. In this paper, we convert the relative clarity problem into a relative blur problem. To prepare the training data for the relative blur problem, we do not use infrared images and only use visible images. We take naturally clear visible images from the COCO 2014 dataset and generate blurred versions of these images by using Gaussian filters with different sizes. The training patch pairs are extracted from the clear images and their corresponding blurred ones. We find that this cross-domain training strategy, which trains in the COCO 2014 dataset and tests in the visible and infrared image fusion dataset, works very well in the IVIF problem.

Specifically, 2200 high-quality, clear images are randomly selected and converted to grayscale images. Then, four blurry versions of each grayscale image is generated by Gaussian filters with four different sizes: $9{\times}9$, $11{\times}11$, $13{\times}13$ and 15×15 (size is 6 times the variance). Both horizontal and vertical derivatives are computed for all images. Then we randomly extracted 1,000,000 patch pairs for training and 300,000 patch pairs for validation. The patch size is 32×32. An example training tuple t consisting of six patches is shown in Fig. 4.

3.3. Relative clarity map generation

After training of our proposed network, it is used to scan each possible pair of visible pixel and infrared pixel by sliding window. For each pair of visible image and infrared image, our network outputs a relative clarity map as shown in Fig. 5. In Fig. 5, the lighter pixel in the relative clarity map means that visible pixel in that position contributes more to fused image than the infrared pixel in the same position.

3.4. Original image enhancement

Due to the poor lighting condition at night, it is better to enhance visible image and infrared image before fusion. We use High Dynamic range compression based on guided filter(GF) [45] to enhance the visible images for night vision. The original visible image I_v with width w and height h is normalized to the range $[0, 255]$. The filtered image I_f is computed as following:

$$I_f = GF_{r,\epsilon}(I_v), \tag{1}$$

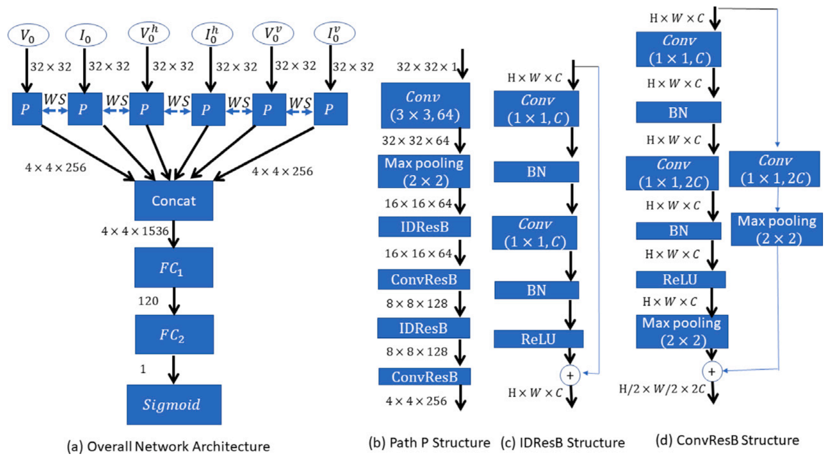

Fig. 3. Our proposed network architecture. (a) Overall network architecture. The input six entry tuple (V^0, I^0, V^h, I^h, V^v, and I^v) is the tuple (visible patch, infrared patch, horizontal derivative of visible patch, horizontal derivative of infrared patch, vertical derivative of visible patch, vertical derivative of infrared patch). Our proposed network contains six paths. Each of the six paths shares the same structure and weights (WS: weight sharing). Each path is denoted by "P". (b) path "P" structure. The path "P" consists of six layers: Convolutional layer, Max Pooling, Identical Residual Block(IDResB), Convolutional Residual Block (ConvResB), Identical Residual Block(IDResB), Convolutional Residual Block (ConvResB). (c) IDResB structure. IDResB has residual structure. It contains five layers: Convolution, Batch Normalization (BN), Convolution, Batch Normalization (BN), and ReLU. IDResB Does not change data dimension. (d) ConvResB structure. ConvResB has residual structure. It has six layers in its main path: Convolution, Batch Normalization (BN), Convolution, Batch Normalization (BN), ReLU, and Pooling. ConvResB has two layers in its skipping path: a convolution layer and a pooling layer. ConvResB reduces spatial dimension to half and doubles channel dimension. The number "$* \times * \times *$" behind each layer is the output data size.

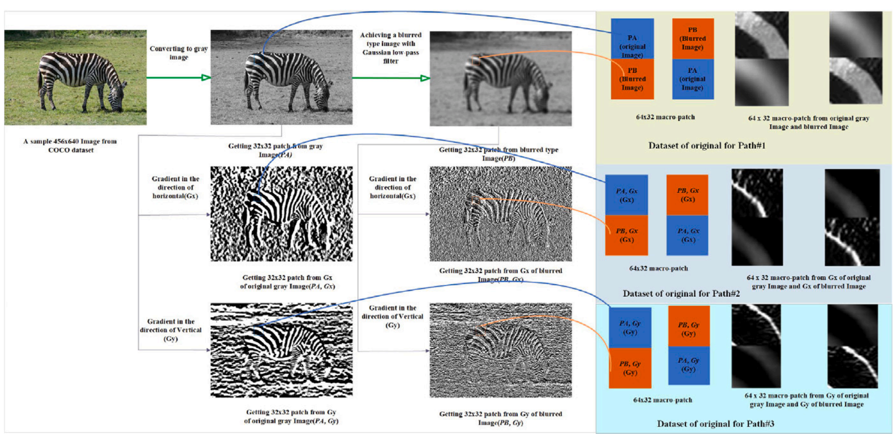

Fig. 4. One training tuple example.

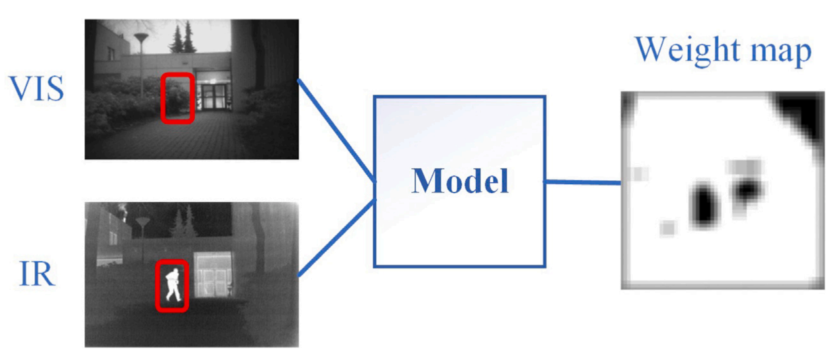

Fig. 5. The schematic of relative clarity generation: The right image is relative clarity map for left input visible and infrared images. The lighter (whiter) pixel in the relative clarity map means that visible pixel contributes more to the fused image than infrared pixel in that position.

where GF is a guided filter (refer to [45] for more information) with filter size $r = \lfloor 0.4 * \max(w, h) \rfloor$ and edge-preserving parameter $\epsilon = 0.01$.

The base layer I_b and detail layer I_d are computed as following [45]:

$$I_b = \log(I_f + \eta) \tag{2}$$

$$I_d = \log(I_v + \eta) - I_b, \tag{3}$$

where log denotes the natural logarithm operator and $\eta = 1$ is used to prevent the log value from being negative.

The dynamic range compression is performed on the base layer I_b with a scaling factor β and the enhanced image I_e is obtained as follows (refer to [46] for more information):

$$I_e = \exp(\beta I_b + I_d + \gamma), \tag{4}$$

Fig. 6. Some examples of enhanced visible images. First row: original visible images; second row: enhanced images.

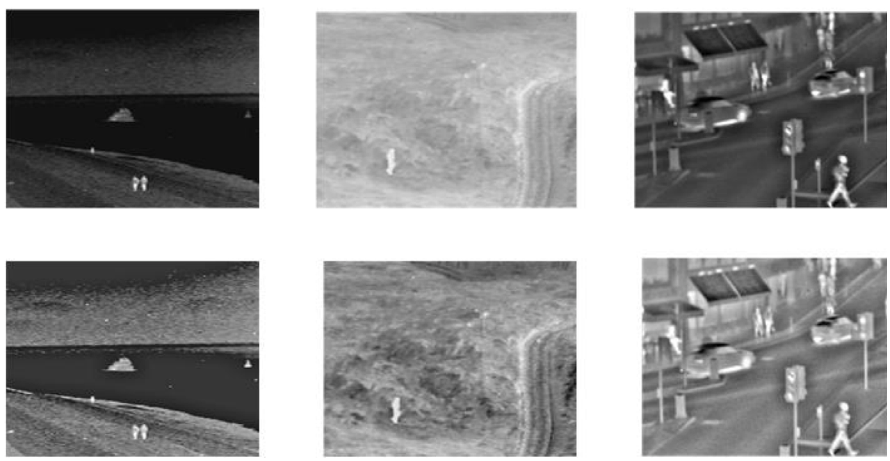

Fig. 7. Some examples of enhanced infrared images. First row: original infrared images; second row: enhanced infrared images.

where $\beta = \frac{\log(T)}{\max(I_b)-\min(I_b)}$, $\gamma = (1-\beta)\max(I_b)$ and the target base contrast T is set to 4. It can be seen that β is less than 1 and that γ is greater than 0. Typically, these settings can considerably enhance the visible image under poor lighting conditions as shown in Fig. 6.

Enhanced version I_i^e of infrared image I_i is obtained by the following linear transformation:

$$I_i^e = \frac{I_i - \min(I_i)}{\max(I_i) - \min(I_i)} \tag{5}$$

Equal illumination can be found in the examplar enhanced infrared images as shown in Fig. 7.

3.5. Fusion method

One can use our relative clarity map from above section to fuse visible and infrared images directly. However, for better fused object contrast, we propose a new fusion method in this paper. First, we threshold clarity map M_c with threshold value 0.5 to obtain binary relative clarity map M_b. Then we compute distance transform map (also called as weight map in this paper) W_1 and W_2 of M_b and $1 - M_b$, respectively. Finally, we use weight map W_1 and W_2 to compute the weighted average of enhanced visible image I_v^e and enhanced infrared image I_i^evisible as final fused image I_{fused} according to the following formula:

$$I_{fused} = \frac{W_1 \cdot I_v^e + W_2 \cdot I_i^e}{W_1 + W_2} \tag{6}$$

Because the center of visible object or infrared object has relatively larger weight, the fused objects have high contrast.

3.6. Enhancement of fused image

We use the single scale retinex (SSR) [47] to further enhance the fused image I_{fused} as following:

$$I_{fused}^e = \log I_{fused} - \log\left(F \circledast I_{fused}\right), \tag{7}$$

Where I_{fused}^e, $\circledast$, and F are the enhanced fusion image, convolution operator and Gaussian filter, respectively. The Gaussian filter F is defined as:

$$F(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(\frac{-\left(x^2 + y^2\right)}{\sigma^2}\right), \tag{8}$$

where (x, y) are pixel coordinates in integer and σ^2 is set to 300 in this paper.

4. Experimental results and analysis

4.1. Training and test details

In dataset preparation, we randomly selected 2200 high-quality clear images from the COCO dataset. As mentioned in Section 3.2, we prepared 1,000,000 patch pairs for training and 300,000 patch pairs for validation. The patch size is 32 × 32.

In the training phase, we select six patches of source images as input. Then we feed to our proposed fully weight shared six-path residual neural network. Each patch was labeled as 1 if it belonged to a positive sample, and 0 otherwise. The experiment was conducted using the following settings: Our computation platform was Linux Ubuntu OS, Intel(R) Core(TM)i7-12700 CPU @ 2.10 GHz, and Nvidia GeForce RTX 2060 GPU. We implemented the model using the PyTorch framework. Stochastic gradient descent (SGD) with a learning rate of 0.0002, momentum of 0.9, and weight decay of 0.0005 was used for training. We applied a step learning rate schedule with a step size of 1 and a gamma value of 0.9. The Cross-Entropy Loss served as the loss criterion, and batch normalization was applied. The training process took approximately 6 h with 30 epochs.

Algorithm 1: Image Fusion based on Multi-Path Residual Network

Training Phase:

1. Randomly select 2200 high-quality clear images from the COCO dataset.
2. Prepare 1,000,000 patch pairs for training and 300,000 patch pairs for validation.
3. Create a fully weight shared six-path residual neural network.
4. Train the network with SGD using the patch pairs and validation pairs (Epochs: 30, LR: 0.0002, Momentum: 0.9, Weight decay: 0.0005).

Test Phase:

1. Select an infrared and visible image pair for testing.
2. Load the pretrained model.
3. Generate a binary classification map.
4. Compute distance transformed maps from the binary classification and its complementary map.
5. Fuse the infrared and visible images using the distance transformed maps.
6. Apply single scale retinex (SSR) for the final fused image.

In the test phase, we select a pair of infrared and visible images as input with the assumption of relative clarity comparison. We solved the comparison problem using a deep binary classification method. Specifically, the tuple $t = (V^0, I^0, V^h, I^h, V^v, I^v)$, consisting of six patches, was labeled as 1 if V^0 was clearer than I^0, and 0 otherwise. This tuple t served as the input for our proposed deep neural network. Then using our pre-trained model, we generate the weight map according to our training strategy. To deal with the discontinuous problem, we compute the two distance transformed maps of the binary classification map and its complementary map. The two distance transformed maps are used as weight maps to fuse the enhanced infrared and visible images. Finally, we use single scale retinex (SSR) to obtain the final fused image.

4.2. Fusion evaluation

The evaluation of image fusion quality continues to be an unresolved problem. This is primarily due to the fact that the same fusion algorithm can yield different results for different types of images. Furthermore, even for the same image, the assessment of fusion effectiveness may vary depending on the specific interests of the observer. Additionally, different fields may have varying requirements for images that contain specific information. To address these challenges, this paper proposes a fusion evaluation method that incorporates both qualitative and quantitative assessment approaches.

4.2.1. Qualitative evaluation

Qualitative assessment in image fusion involves evaluating visual aspects (e.g., color, lightness, fidelity) to determine the satisfaction level of fused images. Mostly, a comparison box is created for direct visual comparison of different fusion algorithms. The fused images are analyzed for color rendering, brightness, and overall fidelity. The assessment enables a comparative analysis of algorithm performance and alignment with human perception. This evaluation is crucial for selecting and refining fusion techniques, ultimately improving image fusion solutions.

4.2.2. Quantitative evaluation

Qualitative evaluation in image fusion is subjective and can vary based on personal preferences. To provide a more quantitative and objective assessment, 11 important evaluation metrics have been introduced for comparison. These metrics offer valuable insights into the performance and quality of the fused images, enabling a more standardized and comprehensive evaluation process.

Among the evaluation metrics used for assessing image fusion, Edge Intensity (EI) and Spatial Frequency (SF) values play a crucial role in measuring the retention of edge and detail information within the fused images [48,49]. Higher EI and SF values signify a greater preservation of these visual elements, contributing to superior fusion results. On the other hand, Entropy (EN) serves to evaluate the information content in the fused image [50]. A larger EN value indicates a higher quantity of retained information, indicating a fusion result of higher quality.

To assess the overall image quality, the metric of Quality of Images (Qab/f) is employed. A larger Qab/f value is indicative of a higher quality fused image [48]. Spatial Frequency Sum of Correlation Coefficients (SCD) measures the correlation between the source images and the fused image [51]. A higher SCD value suggests that a sufficient amount of information from the source images has been preserved, resulting in a fused image that closely resembles the original sources.

Additionally, the evaluation metric of Noise in Images (Nab/f) focuses on the presence of noise in the fused image [52]. A larger Nab/f value signifies a noisier image, potentially impacting the visual quality of the fusion result. Mutual Information between Images (MIab/f) quantifies the shared information between the source images and the fused image [53]. A larger MIab/f value indicates that the fused image retains a significant amount of information from the source images, denoting a successful fusion process.

Moreover, Standard Deviation of Image (SD) and Definition (DF) metrics assess the contrast and sharpness of the fused image [54]. Higher values of SD and DF indicate a higher level of contrast and sharper image, enhancing visual perception and clarity. Visual Information Fidelity (VIFF) metric measures the naturalness and realism of the fused image [55]. A larger VIFF value suggests a more visually appealing fused result with increased authenticity. Lastly, Cross Entropy (CE) [52] evaluates the differences between the source and fused images. A larger CE value signifies a relatively large discrepancy between the two. In summary, larger metric values, with the exception of CE and Nab/f, indicate better fused results in terms of edge and detail preservation, information content, image quality, similarity to source images, contrast, sharpness, and visual fidelity. These metrics provide valuable insights into the quality of infrared and visible image fusion, aiding in the quantitative evaluation and comparison of different fusion techniques.

4.3. Comparison with state of the art methods

In this section, the proposed algorithm is compared to the latest or classic fusion algorithms, including Cross Bilateral Filter (CBF) [56], Sparse Convolutional Representation (ConvSR) [57], Structaware [58], ,HybridMSD [59], DenseFuse [39], NestFuse [37], DDcGan [19], MEF-GAN [38], Image fusion network based on Proportional Maintenance of Gradient and Intensity (PMGI) [60], CNN [11], MDlattLRR [36], Infrared and visible image fusion via gradient transfer and total variation minimization (GTF) [6], Image Fusion Framework based on Convolutional Neural Network (IFCNN) (elementwise-maximum) [35], Unified Unsupervised Image Fusion(U2Fusion) [40], Residual Fusion

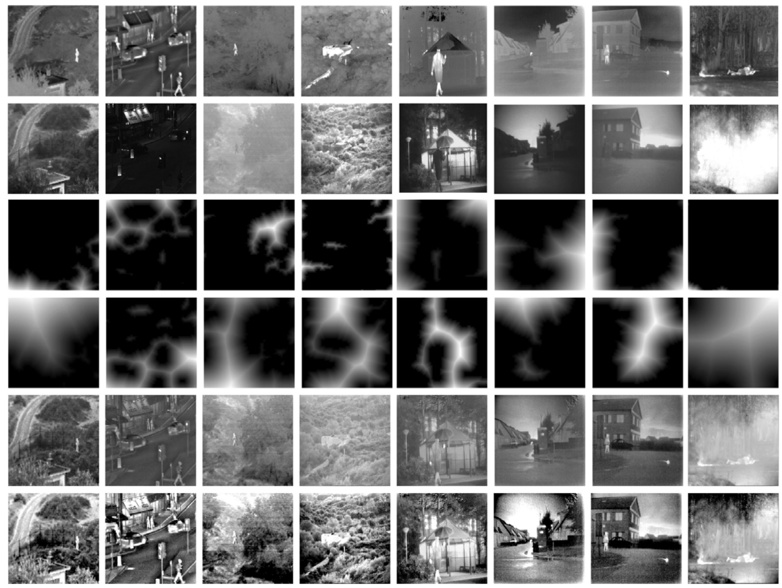

Fig. 8. Example results in TNO dataset: from top row to bottom row: visible input images, infrared input images, distance transformed map W_1, distance transformed map W_2, fused image, and enhanced fusion image.

Network for infrared and visible images(RFN-Nest) [21], Infrared and Visible Image Fusion via Parallel Scene and Texture Learning (PSTLFusion) [61], and Infrared and visible image fusion based on domain transform filtering and sparse representation(DTF-SR) [22]. The results of the other 17 methods are borrowed from their project websites or papers.

In general, these state-of-the-art methods can be categorized into four classes: (1) classical image processing techniques which do not rely on deep learning, such as CBF [56], ConvSR [57], Structaware [58], HybridMSD [59], DTF-SR [22], GTF [6] and MDlattLRR [36]. These methods rely on filters or image decomposition to extract and fuse image features without deep learning. This offers computational efficiency but can struggle to capture complex image details; (2) methods based on auto encoder for deep learning, such as DenseFuse [39], NestFuse [37], and RFN-Nest [21]. These methods leverage autoencoders for feature extraction and fusion in a latent space. While efficient, compressing features can lead to information loss and impact fusion quality. Additionally, their network architecture affects flexibility and generalizability; (3) methods based on the GAN for deep learning, such as DDcGAN [19], FusionGAN [17]and MEF-GAN [38]. These methods employ a generator-discriminator architecture. The generator creates the fused image, while the discriminator guides it towards better results. The lack of ground truth in image fusion makes training challenging, leading to potential convergence issues and inconsistent outcomes. Additionally, understanding their decision-making process is difficult, limiting interpretability and control; (4) deep learning with one-step fusion networks: CNN [11], PMGI [60], PSTLFusion [61] and U2Fusion [40]. These methods, similar to ours, utilize a single deep learning network for direct fusion. While offering comparable performance, they might have less control over the fusion process, limiting fine-tuning for specific needs. Their black-box nature and data dependence can also pose challenges in interpretability and generalizability.

In this paper, our proposed method focuses on weight map generation approaches. Our weight map-based approach employs a deep learning network to generate a weight map before fusing the images. This offers fine-grained control over the fusion process, leading to high accuracy and a simpler training process compared to the aforementioned methods.

4.4. Qualitative results

4.4.1. Visual examplar results across three datasets

Firstly, some examplar results from TNO [62], Roadscene [40], and LLVIP [63] datasets are shown in Fig. 8, Fig. 9 and Fig. 10, respectively. In each of the figure, visible input images, infrared input images, distance transformed map W_1, distance transformed map W_2, fused image, and enhanced fusion image are shown from top row to bottom row, respectively.

From Fig. 8, it can be seen that our fused images have much better quality than the original input visible and infrared images on the top two rows. Salient objects from both infrared and visible images are preserved in fused images. For example, the persons in image 1, 2, 3, 5, 7, and 8 are kept in the final fused images. The house and sky in image 4 and 6 are also preserved in final fused images.

From Fig. 9, it can be seen that both target information from infrared images and background information from visible images are preserved in our fused images. Our fused images are more pleasant to human eyes than the input visible and infrared images.

From Fig. 10, it can be seen again that our fused images successfully preserve salient information from the infrared domain, while simultaneously retaining texture details from visible objects.

In summary, it can be seen that our method can achieve consistent good fusion results over multiple datasets. The salient objects in both infrared and visible images can be preserved accurately.

4.4.2. Visual comparison examples

To conduct a qualitative comparison of our method with others, we have chosen images from three distinct datasets: the TNO dataset, the road scene dataset, and the LLVIP dataset. Specifically, Figs. 11, and 12 are selected from the TNO dataset, Fig. 13 is chosen from the road scene dataset, and Fig. 14 is selected from the LLVIP dataset. To facilitate comparison, we present the results of 17 other methods alongside our method in Figs. 11, 12, 13, and 14.

From Fig. 11, it can be seen that our fused image outperforms the fused images produced by other methods. The "person with backpack" in the green box is kept in all methods including ours. However, the "table and stand chairs" in the yellow box are more clear in our fused image than in the fused images from all other methods. The "house" in the blue box show much details in our fused image. However, the details of the "house" are lost in other fused images from other

Fig. 9. Example results in Roadscene dataset: from top row to bottom row: visible input images, infrared input images, distance transformed map W_1, distance transformed map W_2, fused image, and enhanced fusion image.

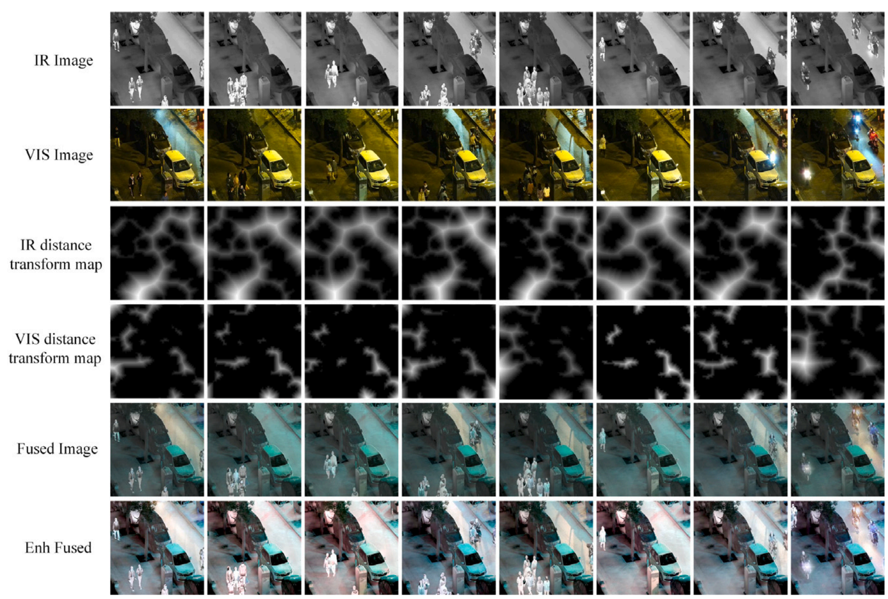

Fig. 10. Example results in LLVIP dataset: from top row to bottom row: visible input images, infrared input images, distance transformed map W_1, distance transformed map W_2, fused image, and enhanced fusion image.

methods. For example, the windows in the top left of the blue box are clear in our fused image and not clear in other fused images from other methods. Our fused image appearance is pleasant for human eyes.

From Fig. 12, it can be seen that our fused image has high contrast and many details from the input visible image. The tree in the blue box is clearer in our fused image than in other fused images. Although, the "garden lamp" in yellow box is kept in all methods. Our reserved "garden lamp" has a better shape. The "outside bench" in the green box is kept well in all fused images. And also from Fig. 13, it is evident that most methods can retain the details of "wall text and number" (highlighted in the red box) and the "white-painted tree root" (highlighted in the yellow box). However, our proposed method demonstrates precise preservation of information related to "wall text and number" and "white-painted tree root". In contrast, methods such as ConvSR, FusionGAN, and PSTLFusion exhibit loss in the "number on the wall", and additional methods including CBF, ConvSR, Structure Aware, PSTLFusion, and DTF-SR also fail to preserve the "white-painted tree root". Importantly, our method does not introduce any noticeable artifacts into the fused result, and it accurately preserves the background.

In Fig. 14, our fused result excels when compared to other methods in terms of both appearance and detailed texture information. The "car" highlighted in the red box remains well preserved across all methods. However, certain methods, such as CBF, introduce artifacts into the fused result, while others like ConvSR, Hybrid-MSD, FusionGAN, DDcGAN, PMGI, U2fusion, RFN-Nest, and PSTLFusion exhibit blurred results in the red box. In contrast, our proposed method accurately preserves the "car" in the red box.

In general, our proposed fusion method preserves much visible details and infrared objects. The overall appearance of our fused images are pleasant for human eyes.

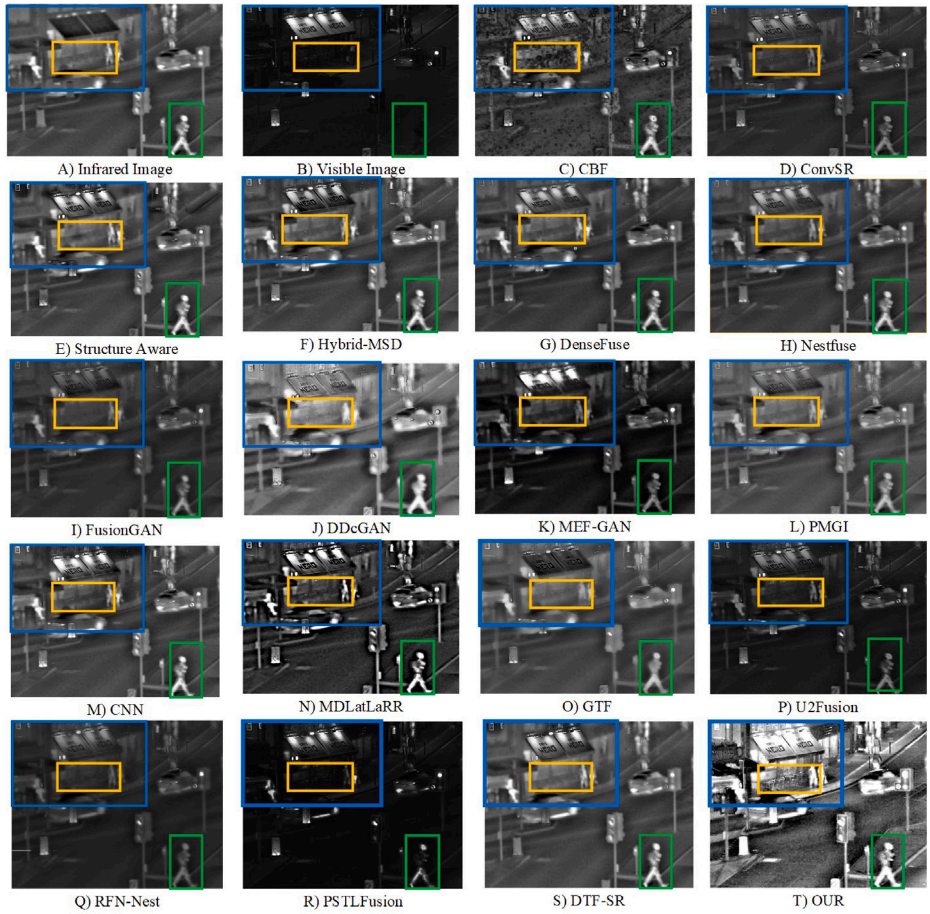

Fig. 11. Comparative results in the "man on the street" image. Yellow box: table and stand chairs ; green box: person with backpack; blue box: house.

Table 1
Quantitative comparative results in TNO datasets: The best, second-best, and third-best values in the tables are highlighted in red, green, and blue colors, respectively. ↑: larger is better; ↓: smaller is better.

Methods	EI ↑	CE ↓	SF ↑	EN ↑	Qab/f ↑	SCD ↑	Nab/f ↓	MIab/f ↑	SD ↑	DF ↑	VIFF ↑
CBF [56]	52.3045	1.3163	13.0921	6.8275	0.4539	1.3193	0.0621	13.655	34.6491	6.4624	0.2834
ConvSR [57]	46.0165	0.8151	16.1643	7.0725	0.5003	1.0409	0.0427	14. 1451	45.1763	5.6174	0.2946
StructAware [58]	44.9083	0.9165	11.1984	7.0549	0.6228	1.2611	0.0147	14.1098	40.3137	5.4062	0.3524
HybridMSD [59]	46.32	1.3157	12.3459	6.7103	0.5014	1.5773	0.0435	13.4206	31.2742	6.0954	0.4448
DenseFuse [39]	36.4838	1.4015	9.3238	6.8526	0.4744	1.5329	0.0352	13.7053	38.0412	4.6176	0.3876
NestFuse [37]	38.4401	1.4458	9.7098	6.8856	0.5002	1.5839	0.0491	13.7713	38.3311	4.9099	0.3744
FusionGAN [17]	32.5997	1.9353	8.0476	6.5409	0.2685	0.6876	0.0352	13.0817	29.1495	4.2727	0.1473
DDcGAN [19]	54.8233	1.0979	12.8351	7.4215	0.3613	1.3831	0.1323	14.8429	46.7319	6.9395	0.4634
MEF-GAN [38]	36.905	1.498	7.8481	6.9727	0.2076	1.3083	0.075	13.9454	43.7332	3.8742	0.324
PMGI [60]	37.2133	1.5656	8.7194	6.8688	0.379	1.5738	0.034	13.7376	33.0167	4.4328	0.4237
CNN [11]	44.8334	0.8939	11.6483	7.0629	0.5833	1.6006	0.0234	14. 1259	44.9159	5.7266	0.4504
MDLatLRR [36]	86.8855	1.0159	21.6245	7.0897	0.3238	1.5641	0.3021	14.1793	43.0009	11.0665	0.8776
GTF [6]	32.9905	1.1811	8.7222	6.6375	0.4146	0.9629	0.0150	13.2750	31.5768	4.1373	0.1876
U2Fusion [40]	48.4915	1.3255	11.0368	6.7227	0.3937	1.5946	0.08	13.4453	31.3794	5.8343	0.5331
RFN-Nest [21]	29.1473	1.6651	6.1274	6.8413	0.3359	1.5047	0.0162	13.6826	35.2704	3.0704	0.4755
PSTLFusion [61]	39.6384	1.3695	10.6651	6.8862	0.3629	1.3201	0.0546	13.7724	43.8088	5.0044	0.4435
DTF-SR [22]	41.0071	0.9539	10.4648	7.1605	0.5960	1.2393	0.0157	14.3211	43.1978	4.9692	0.3636
Ours	91.8262	1.2329	21.2759	7.7437	0.2593	1.4714	0.3217	15.4873	68.1418	11.0018	1.3533

4.5. Quantitative comparison

In this section, for a quantitative comparison, we present comparative metric values for various datasets: Table 1 for the TNO dataset, Table 2 for the roadscene dataset, and Table 3 for the LLVIP dataset. The best, second-best, and third-best values in the tables are highlighted in red, green, and blue colors, respectively.

From Table 1, it can be seen that our proposed method performances best in six metrics (EI, SF, EN, MI, SD, and VIFF) and secondly in one metrics (DF) among all 18 methods. In Table 2, our proposed methods outperformed among all 18 methods by achieving the top three metrics (SCD, SD, and VIFF), the second-best scores in three metrics (EI, SF, and EN), and the third-best score in one metric (DF). Moving on to Table 3, where we conducted a qualitative analysis specifically on the LLVIP dataset, our proposed methods secured the top position in five metrics (EN, SF, EN, MI, and DF) and the second-best scores in three metrics (SCD, SD, and VIFF) among all 18 methods.

Significant values of EI, SF, EN, and MI highlight the proficiency of our method in extracting edge information and detailed content from source images. The highest VIFF value suggests that our fused image

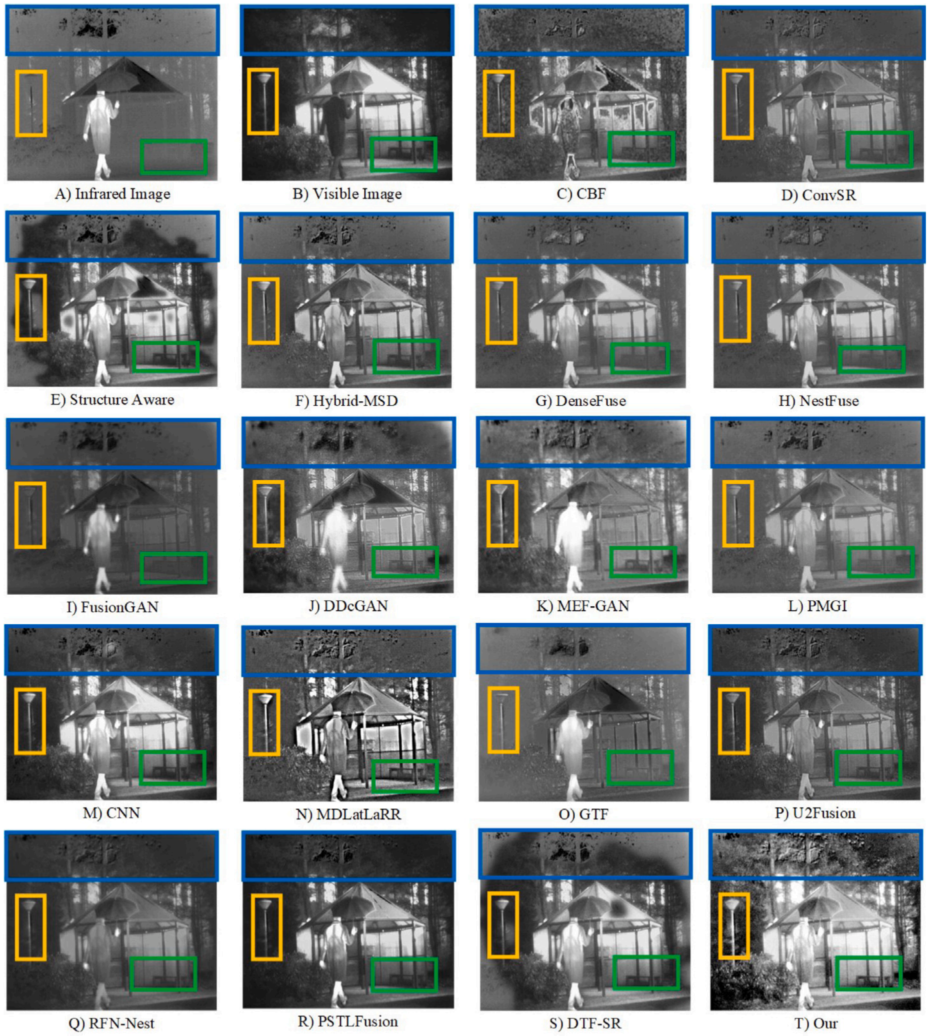

Fig. 12. Comparative results in the "umbrella man" image. Blue box: trees; yellow box: garden lamp; green box: outside bench.

Table 2
Quantitative comparative results in roadscene datasets: The best, second-best, and third-best values in the tables are highlighted in red, green, and blue colors, respectively. ↑: larger is better; ↓: smaller is better.

Methods	EI ↑	CE ↓	SF ↑	EN ↑	Qab/f ↑	SCD ↑	Nab/f ↓	MIab/f ↑	SD ↑	DF ↑	VIFF ↑
CBF [56]	67.6171	0.8217	16.4106	7.53223	0.49706	0.8784	0.03878	15.0645	51.982	7.99372	0.35935
ConvSR [57]	52.3536	1.4035	13.8076	7.33251	0.47517	1.17765	0.032	14.665	50.3443	6.23511	0.47627
StructAware [58]	60.1213	0.66226	15.0096	7.45623	0.61826	0.7932	0.01876	15.3125	57.559	6.8959	0.39855
HybridMSD [59]	62.6781	1.07577	16.4785	7.03235	0.52717	1.27241	0.04733	14.0647	37.0965	7.54094	0.47967
DenseFuse [39]	50.5087	1.67749	13.0556	7.20229	0.46529	1.23979	0.01651	14.4046	44.1531	6.21341	0.55629
NestFuse [37]	54.6225	1.33365	14.5847	7.38199	0.49101	1.24737	0.04956	14.764	51.6212	6.35414	0.4734
FusionGAN [17]	36.5697	1.96201	9.14531	7.1822	0.13094	0.41629	0.04149	14.3644	42.3497	4.19561	0.08502
DDcGAN [19]	53.4136	1.28741	13.2014	7.58324	0.34251	1.19864	0.06258	15.1665	58.1717	5.95045	0.48036
MEF-GAN [38]	38.1304	1.53924	9.40596	6.40821	0.17763	1.09448	0.06368	12.8164	51.0338	3.85455	0.22984
PMGI [60]	47.2067	1.63954	10.9368	7.34928	0.42481	1.09892	0.01458	14.6986	49.3262	5.12879	0.45184
CNN [11]	58.5654	0.89004	15.0229	7.27116	0.57479	1.40242	0.02793	14.5423	44.9717	6.93283	0.47032
MDLatLRR [36]	81.2228	1.00825	20.6882	7.18656	0.49099	1.43029	0.11149	14.3731	42.583	9.39409	0.70745
GTF [6]	37.8936	0.82046	10.2563	7.63903	0.37306	0.80678	0.0117	15.2781	59.7686	4.42761	0.24814
U2Fusion [40]	66.1014	1.52487	16.2073	7.19923	0.47774	1.35491	0.06752	14.3985	42.9839	7.73035	0.54705
RFN-Nest [21]	38.4507	1.62098	8.33478	7.33936	0.31717	1.23271	0.01235	14.6787	49.469	3.95848	0.5129
PSTLFusion [61]	49.595	1.74575	12.7143	6.97766	0.2197	0.43251	0.06948	13.9553	39.7794	5.76714	0.0005
DTF-SR [22]	56.3857	0.66047	14.4484	7.65257	0.39394	0.7857	0.01113	15.3051	57.541	6.63301	0.39687
Ours	71.1484	0.97105	18.0142	7.58727	0.37697	1.49808	0.15968	15.1745	72.1435	7.75837	0.95385

achieves the most realistic and natural appearance. Additionally, the relatively large DF and SD values indicate a high contrast in the fused image. In summary, the pipeline proposed in this study demonstrates superior performance in both quantitative and qualitative aspects when compared with other state-of-the-art methods.

4.6. Ablation study

In this study, we conducted three ablation studies to showcase the effectiveness of our proposed method in image fusion. To evaluate its performance, experiments were carried out on three datasets:

Fig. 13. Comparative results from roadscene dataset. Yellow box: white painted tree root ; red box: wall text and number.

Table 3
Quantitative comparative results in LLVIP datasets: The best, second-best, and third-best values in the tables are highlighted in red, green, and blue colors, respectively. ↑: larger is better; ↓: smaller is better.

Methods	EI ↑	CE ↓	SF ↑	EN ↑	Qab/f ↑	SCD ↑	Nab/f ↓	MIab/f ↑	SD ↑	DF ↑	VIFF ↑
CBF [56]	61.1751	0.5856	17.098	7.2469	0.7057	0.8164	0.0161	14.4937	47.1519	6.9482	0.5507
ConvSR [57]	39.0759	0.803	14.098	7.2198	0.6713	1.3691	0.0035	14.4396	43.5004	4.6283	0.4783
StructAware [58]	47.1894	0.5258	15.461	7.3407	0.7612	0.8351	0.009	14.6814	50.6289	5.3094	0.6114
HybridMSD [59]	46.5333	0.601	15.787	7.2623	0.7428	1.1134	0.0146	14.5246	51.6514	5.3118	0.6487
DenseFuse [39]	27.3023	0.9165	8.9779	7.1952	0.4009	1.386	0.001	14.3905	42.9304	3.3106	0.3972
NestFuse [37]	23.7768	0.8663	9.5782	7.3835	0.4089	1.355	0.0003	14.767	48.8278	2.6434	0.3633
FusionGAN [17]	20.522	2.7382	7.0321	6.7604	0.1286	0.4737	0.0129	13.5209	31.143	2.4496	0.1012
DDcGAN [19]	38.332	1.2426	12.491	7.422	0.5242	1.4767	0.0117	14.8439	52.6266	4.5357	0.4076
MEF-GAN [38]	30.2201	0.7544	9.5306	7.5301	0.301	1.7002	0.0227	15.0602	84.3323	3.1266	0.5203
PMGI [60]	23.8467	1.8295	7.3952	7.041	0.3087	1.0525	0.0015	14.0821	36.7464	2.725	0.236
CNN [11]	59.3762	0.5745	20.948	7.43	0.5683	1.3615	0.0194	14.86	57.7748	7.9156	0.5667
MDLatLRR [36]	71.9014	0.6211	24.007	7.4497	0.5615	1.364	0.1005	14.8993	53.4169	8.1658	0.9028
GTF [6]	34.2801	0.6401	12.515	7.4672	0.5055	1.0092	0.0064	14.9344	51.6667	4.1818	0.2859
U2Fusion [40]	40.1356	0.7272	12.468	7.238	0.619	1.515	0.0099	14.476	49.5516	4.5853	0.5215
RFN-Nest [21]	25.2062	0.7479	6.8626	7.2756	0.3185	1.4933	0.0017	14.5512	45.5391	2.7062	0.3918
PSTLFusion [61]	17.7887	3.8757	6.9389	5.8789	0.1138	0.1829	0.0137	11.7577	16.5742	2.1751	0.051
DTF-SR [22]	45.6299	0.4997	15.345	7.3817	0.7537	0.8535	0.007	14.7634	51.9053	5.1942	0.6196
Ours	74.6603	0.9458	24.087	7.8639	0.3758	1.6486	0.0782	15.7277	69.743	10.277	0.709

TNO, Roadmap, and LLVIP. The primary focus was on assessing the impact of a post-processing strategy known as single scale retinex. Both qualitative analysis, illustrated in Fig. 16, and quantitative assessments, detailed in Table 4, were performed. Fig. 15 visually represents the ablation studies, with the first ablation showing the direct model fused result, the second demonstrating the result after applying the guided filter, and the third exhibiting the result after single scale retinex enhancement. Quantitative analysis, presented in Table 4, further highlights the superiority of ablation 3 over other cases.

Upon analyzing the results in Fig. 16 and Table 4, it becomes evident that the guided filter, when applied after the merging process, does not significantly influence the improvement in the resulting image's quality. This is attributed to our approach of enhancing the visible image before merging it with the weight map. Conversely, the proposed

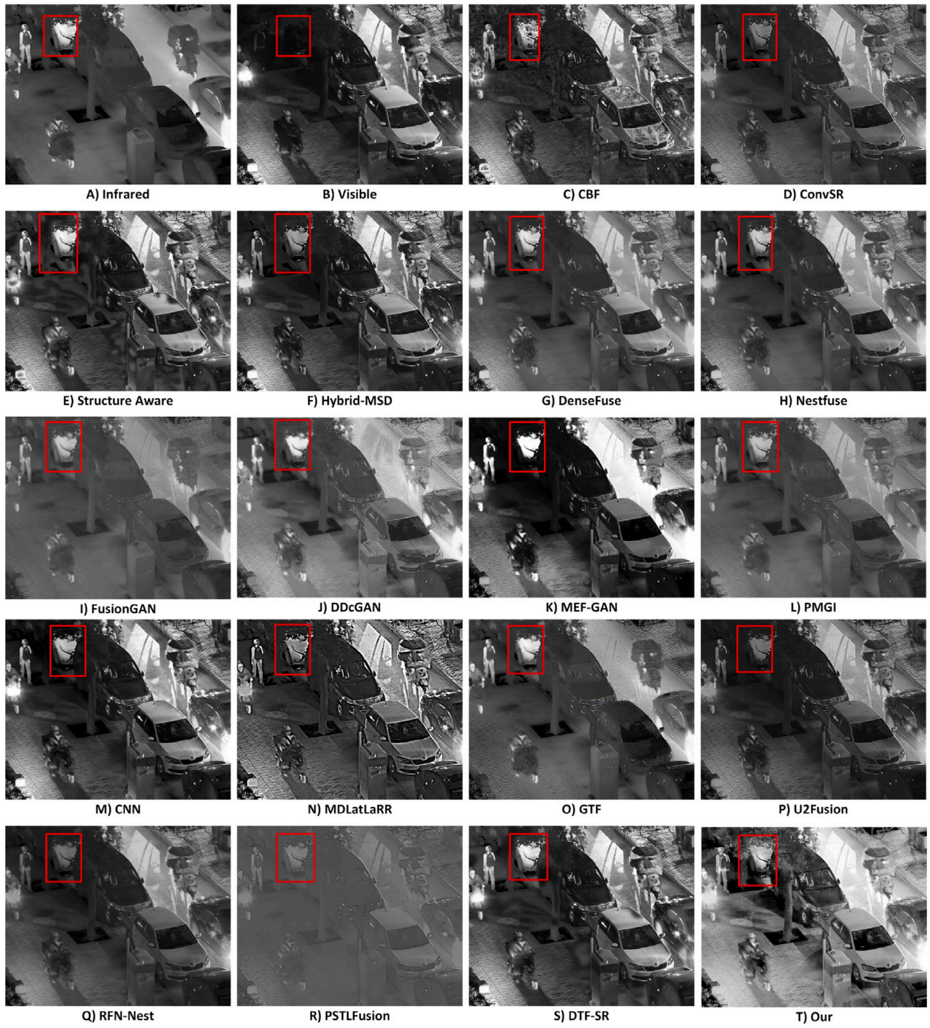

Fig. 14. Comparative results from LLVIP dataset. red box: white car.

Table 4
Objective evaluation results of our proposed method in TNO, RoadScene and LLVIP datasets with/without guided filter(GF), with/without Single Scale Retinex(SSR) step. ↑: larger is better; ↓: smaller is better.

Model output	Dataset	EI ↑	CE ↓	SF ↑	EN ↑	Qab/f ↑	SCD ↑	Nab/f ↓	MIab/f ↑	SD ↑	DF ↑	VIFF ↑
Direct model fused result	TNO	19.79503945	1.36968928	4.598762406	6.261635923	0.243816665	1.563651454	0.001067624	12.52327185	23.89145116	2.098477296	0.215542295
	Roadscene	24.14227487	0.860335331	5.413964128	6.668286806	0.293832804	1.313800482	0.000671801	13.33657361	29.64941461	2.493759422	0.291440686
	LLVIP	27.24898321	0.853329386	8.370262095	7.214483167	0.250040811	1.261677787	0.001358155	14.42896633	42.35039112	3.087226288	0.293927932
Fused result after guided filter enhancement	TNO	40.66358513	2.72198437	9.40347564	6.69765701	0.31631881	1.56131734	0.08603091	13.39531403	28.42222828	4.60453665	0.54653237
	Roadscene	45.47671995	1.36392741	10.51802227	7.07448724	0.40976070	1.40832160	0.03750781	14.14897449	37.83240572	4.92966624	0.51609532
	LLVIP	46.64342200	2.77572692	14.03272006	7.24593560	0.34609850	0.94789820	0.01688366	14.49187121	38.82950407	6.16149585	0.40826299
Fused result after single scale retinex enhancement	TNO	91.82618745	1.232935843	21.27597733	7.74366728	0.259287707	1.471450842	0.321673671	15.48733456	68.14179963	11.00177749	1.353296868
	Roadscene	71.14838247	0.971046864	18.01424622	7.587265831	0.376974364	1.49807549	0.159679521	15.17453166	72.14348187	7.758371578	0.953853268
	LLVIP	74.6602791	0.945823152	24.08716656	7.863860192	0.375842238	1.648552418	0.078239892	15.72772038	69.74295928	10.2765688	0.708992344

Single Scale Retinex stands out, making a substantial difference in enhancing the quality of the resultant image, as demonstrated in Fig. 16 and Table 4.

The outcomes showcased in Fig. 16 clearly indicate that implementing single scale retinex as a post-processing step improves the fused image compared to images obtained before post-processing. The enhanced images exhibit improved visual appearance and better preservation of detail information. To validate the efficacy of this post-processing approach, objective evaluations were conducted on three datasets, as detailed in Table 4. The results in Table 4 confirm an improvement in most evaluation metrics. Higher metric values, except for CE and Nab/f, indicate better quality of the fused images. However, we observed some issues with certain quantitative metrics in Table 4, specifically Noise in Images (Nab/f) and Quality of Image (Qab/f), after applying the post-processing. The single scale retinex enhancement introduces noise and excessive brightness in some lighter areas of the fused image, adversely affecting its quality. For instance, in the fourth column of Fig. 16, a significant amount of noise is visible in the fusion image. Similarly, in the fifth column of Fig. 16, the window details of the fusion image are weakened. This demonstrates that the brighter areas of the image are susceptible to negative effects from the single scale retinex process.

However, it is important to note that overall, our proposed post-processing technique, single scale retinex, excels at extracting detail targets, preserving detail information, maintaining visual elements, establishing correlation between source and fused images, as well as retaining edge and detail information within the fused images. Despite

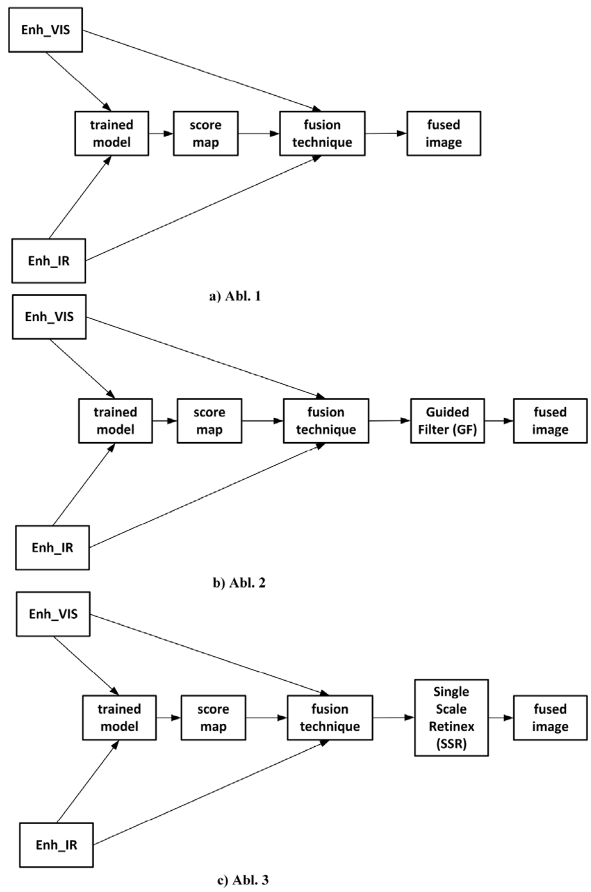

Fig. 15. Ablation study experiments. (a), direct model fused result; (b), fused result after guided filter enhancement (c), fused result after single scale retinex enhancement.

the challenges observed in specific scenarios, the benefits of single scale retinex are substantial.

4.7. Comparison of computational efficiency

In this section, we performed a computational efficiency comparison by computing the average execution times of 21 pairs of infrared and visible image fusions. To compare the time complexity, we categorized state-of-the-art methods in two classes: (1) classical image processing techniques which do not rely on deep learning, such as CBF [56], ConvSR [57], Structaware [58], HybridMSD [59], DTF-SR [22], GTF [6] and MDlattLRR [36]. (2) Methods based on deep learning. This is the main comparison method of our method such as CNN [11], PMGI [60], PSTLFusion [61], DenseFuse [39], NestFuse [37], FusionGAN [17], DDcGAN [19], MEF-GAN [38], RFN-Nest [21]. and U2Fusion [40].

In Table 5, we analyze their inference time complexity and find that the proposed method is 14th out of 18th in terms of inference time efficiency. One of the reasons for the poor inference time of our method is the utilization of a sliding-window technique during the reconstruction process. This sliding window approach adds complexity to the inference phase, as it involves processing the input data in smaller overlapping segments. Additionally, our method employs post-processing steps, which further contribute to the overall inference time. However, it is worth noting that during the training phase, our proposed method exhibits fast convergence. In fact, it achieves convergence within just two epochs, and the entire training process takes less than one hour on NVIDIA GetForce RTX 2060. This indicates that the training time for our method is relatively efficient and does not pose a significant burden. In future work, we intend to optimize the reconstruction time by adopting an end-to-end process, thus enhancing the overall efficiency of our method.

5. Conclusion

In this paper, a new method is proposed for IVIF problem. Specifically, visible images are enhanced by guided filter and high dynamic range compression. Infrared images are normalized by linear transformation. Then we use blur and clear discrimination to detect salient pixels between infrared and visible images. A fully weight shared multi-path residual neural network is proposed for blur discrimination between infrared and visible image pixels in the same position. Clear pixels are treated as salient pixels which contribute to fused images more than blur pixels. The output of our proposed network is a binary classification map for blur and clear discrimination and is treated as our fusion weight map in fusion stage. To deal with the discontinuous problem, we compute the two distance transformed maps of the binary classification map and its complementary map. The two distance transformed maps are used as weight map to fuse the enhanced infrared and visible images. Finally, we use single scale retinex to further enhance our fused images. The experimental results in public IVIF datasets demonstrate the superior performance of our proposed approach over other state-of-the-art methods in terms of both subjective visual quality and objective metrics.

CRediT authorship contribution statement

Deboch Eyob Abera: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Jin Qi:** Writing – review & editing, Supervision, Funding acquisition. **Jian Cheng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Table 5
Comparison of average running times for 21 pair images set up on NVIDIA GetForce RTX 2060: The tables highlight the best, second-best, and third-best values in red, green, and blue colors, respectively.

Classical Methods	CBF	ConvSR	StructAware	HybridMSD	PMGI	MDLatLRR	GTF	DTF-SR		
Times/s	400.575595	4378.014871	7.955524	118.945431	10.340032	2686.499965	7.751941	1869.093451		
Deep learning Methods	CNN	FusionGAN	PSTLFusion	MEF-GAN	NestFuse	Dense Fuse	RFN-Nest	DDcGAN	U2Fusion	Our
Times/s	1570.32677	42.46748	26.53001	81.119691	8.191057	570.37286	10.563031	48.02515	27.582664	589.453703

Fig. 16. Visualization ablation experiment: from top bottom row: input visible images, input infrared images,direct model fused result, fused result after guided filter enhancement, fused result after single scale retinex enhancement.

References

[1] J. Ma, Y. Ma, C. Li, Infrared and visible image fusion methods and applications: A survey, Inf. Fusion 45 (2019) 153–178.

[2] C. Sun, C. Zhang, N. Xiong, Infrared and visible image fusion techniques based on deep learning: A review, Electronics 9 (12) (2020) 2162.

[3] L. Zhan, Y. Zhuang, L. Huang, Infrared and visible images fusion method based on discrete wavelet transform, J. Comput. 28 (2) (2017) 57–71.

[4] B.S. Kumar, Multifocus and multispectral image fusion based on pixel significance using discrete cosine harmonic wavelet transform, Signal, Image Video Process. 7 (6) (2013) 1125–1143.

[5] J. Ma, C. Chen, C. Li, J. Huang, Infrared and visible image fusion via gradient transfer and total variation minimization, Inf. Fusion 31 (2016) 100–109.

[6] J. Ma, C. Chen, C. Li, J. Huang, Infrared and visible image fusion via gradient transfer and total variation minimization, Inf. Fusion 31 (2016) 100–109.

[7] Y. Liu, Z. Wang, Simultaneous image fusion and denoising with adaptive sparse representation, IET Image Process. 9 (5) (2015) 347–357.

[8] H. Li, X.-J. Wu, J. Kittler, MDLatLRR: A novel decomposition method for infrared and visible image fusion, IEEE Trans. Image Process. 29 (2020) 4733–4746.

[9] S. Yin, L. Cao, Q. Tan, G. Jin, Infrared and visible image fusion based on NSCT and fuzzy logic, in: 2010 IEEE International Conference on Mechatronics and Automation, IEEE, 2010, pp. 671–675.

[10] H. Luo, R. Hou, W. Qi, A NSP-based dictionary learning for image fusion, in: 2018 Chinese Control and Decision Conference, CCDC, IEEE, 2018, pp. 4831–4834.

[11] Y. Liu, X. Chen, H. Peng, Z. Wang, Multi-focus image fusion with a deep convolutional neural network, Inf. Fusion 36 (2017) 191–207.

[12] H. Li, X.-J. Wu, J. Kittler, Infrared and visible image fusion using a deep learning framework, in: 2018 24th International Conference on Pattern Recognition, ICPR, IEEE, 2018, pp. 2705–2710.

[13] Y. Chen, H. Xie, H. Shin, Multi-layer fusion techniques using a CNN for multispectral pedestrian detection, IET Comput. Vis. 12 (8) (2018) 1179–1187.

[14] Y. Liu, X. Chen, J. Cheng, H. Peng, Z. Wang, Infrared and visible image fusion with convolutional neural networks, Int. J. Wavelets Multiresolut. Inf. Process. 16 (03) (2018) 1850018.

[15] H. Li, X. jun Wu, T.S. Durrani, Infrared and visible image fusion with ResNet and zero-phase component analysis, Infrared Phys. Technol. 102 (2019) 103039.

[16] X. Song, X.-J. Wu, Multi-focus image fusion with PCA filters of PCANet, in: F. Schwenker, S. Scherer (Eds.), Multimodal Pattern Recognition of Social Signals in Human-Computer-Interaction, Springer International Publishing, 2019, pp. 1–17.

[17] J. Ma, W. Yu, P. Liang, C. Li, J. Jiang, FusionGAN: A generative adversarial network for infrared and visible image fusion, Inf. Fusion 48 (2019) 11–26.

[18] J. Ma, P. Liang, W. Yu, C. Chen, X. Guo, J. Wu, J. Jiang, Infrared and visible image fusion via detail preserving adversarial learning, Inf. Fusion 54 (2020) 85–98.

[19] J. Ma, H. Xu, J. Jiang, X. Mei, X.-P. Zhang, DDcGAN: A dual-discriminator conditional generative adversarial network for multi-resolution image fusion, IEEE Trans. Image Process. 29 (2020) 4980–4995.

[20] K.R. Prabhakar, V.S. Srikar, R.V. Babu, DeepFuse: A deep unsupervised approach for exposure fusion with extreme exposure image pairs, in: 2017 IEEE International Conference on Computer Vision, ICCV, 2017, pp. 4724–4732.

[21] H. Li, X.-J. Wu, J. Kittler, RFN-nest: An end-to-end residual fusion network for infrared and visible images, Inf. Fusion 73 (2021) 72–86.

[22] X. Li, H. Tan, F. Zhou, G. Wang, X. Li, Infrared and visible image fusion based on domain transform filtering and sparse representation, Infrared Phys. Technol. 131 (2023) 104701.

[23] Q. Li, W. Wu, L. Lu, Z. Li, A. Ahmad, G. Jeon, Infrared and visible images fusion by using sparse representation and guided filter, J. Intell. Transp. Syst. 24 (3) (2020) 254–263.

[24] J. Wang, J. Peng, X. Feng, G. He, J. Fan, Fusion method for infrared and visible images by using non-negative sparse representation, Infrared Phys. Technol. 67 (2014) 477–489.

[25] Q. Zhang, Y. Liu, R.S. Blum, J. Han, D. Tao, Sparse representation based multi-sensor image fusion for multi-focus and multi-modality images: A review, Inf. Fusion 40 (2018) 57–75.

[26] D.P. Bavirisetti, G. Xiao, G. Liu, Multi-sensor image fusion based on fourth order partial differential equations, in: 2017 20th International Conference on Information Fusion, Fusion, IEEE, 2017, pp. 1–9.

[27] W. Kong, Y. Lei, H. Zhao, Adaptive fusion method of visible light and infrared images based on non-subsampled shearlet transform and fast non-negative matrix factorization, Infrared Phys. Technol. 67 (2014) 161–172.

[28] X. Zhang, Y. Ma, F. Fan, Y. Zhang, J. Huang, Infrared and visible image fusion via saliency analysis and local edge-preserving multi-scale decomposition, J. Opt. Soc. Amer. A 34 (8) (2017) 1400–1410.

[29] J. Zhao, Y. Chen, H. Feng, Z. Xu, Q. Li, Infrared image enhancement through saliency feature analysis based on multi-scale decomposition, Infrared Phys. Technol. 62 (2014) 86–93.

[30] T. Xiang, L. Yan, R. Gao, A fusion algorithm for infrared and visible images based on adaptive dual-channel unit-linking PCNN in NSCT domain, Infrared Phys. Technol. 69 (2015) 53–61.

[31] W. Kong, L. Zhang, Y. Lei, Novel fusion method for visible light and infrared images based on NSST–SF–PCNN, Infrared Phys. Technol. 65 (2014) 103–112.

[32] Y. Liu, S. Liu, Z. Wang, A general framework for image fusion based on multi-scale transform and sparse representation, Inf. Fusion 24 (2015) 147–164.

[33] J. Ma, Z. Zhou, B. Wang, H. Zong, Infrared and visible image fusion based on visual saliency map and weighted least square optimization, Infrared Phys. Technol. 82 (2017) 8–17.

[34] J. Zhao, G. Cui, X. Gong, Y. Zang, S. Tao, D. Wang, Fusion of visible and infrared images using global entropy and gradient constrained regularization, Infrared Phys. Technol. 81 (2017) 201–209.

[35] Y. Zhang, Y. Liu, P. Sun, H. Yan, X. Zhao, L. Zhang, IFCNN: A general image fusion framework based on convolutional neural network, Inf. Fusion 54 (2020) 99–118.

[36] H. Li, X.-J. Wu, J. Kittler, MDLatLRR: A novel decomposition method for infrared and visible image fusion, IEEE Trans. Image Process. 29 (2020) 4733–4746.

[37] H. Li, X.-J. Wu, T. Durrani, NestFuse: An infrared and visible image fusion architecture based on nest connection and spatial/channel attention models, IEEE Trans. Instrum. Meas. 69 (12) (2020) 9645–9656.

[38] H. Zhang, Z. Le, Z. Shao, H. Xu, J. Ma, MEF-GAN: An unsupervised generative adversarial network with adaptive and gradient joint constraints for multi-focus image fusion, Inf. Fusion 66 (2021) 40–53.

[39] H. Li, X.-J. Wu, DenseFuse: A fusion approach to infrared and visible images, IEEE Trans. Image Process. 28 (5) (2018) 2614–2623.

[40] H. Xu, J. Ma, J. Jiang, X. Guo, H. Ling, U2Fusion: A unified unsupervised image fusion network, IEEE Trans. Pattern Anal. Mach. Intell. (2020).

[41] J. Ma, L. Tang, F. Fan, J. Huang, X. Mei, Y. Ma, SwinFusion: Cross-domain long-range learning for general image fusion via swin transformer, IEEE/CAA J. Autom. Sin. 9 (7) (2022) 1200–1217.
[42] L. Tang, Y. Deng, Y. Ma, J. Huang, J. Ma, SuperFusion: A versatile image registration and fusion network with semantic awareness, IEEE/CAA J. Autom. Sin. 9 (12) (2022) 2121–2137.
[43] Y. Fu, X.-J. Wu, T. Durrani, Image fusion based on generative adversarial network consistent with perception, Inf. Fusion 72 (2021) 110–125.
[44] M. Amin-Naji, A. Aghagolzadeh, M. Ezoji, Ensemble of CNN for multi-focus image fusion, Inf. Fusion 51 (2019) 201–214.
[45] F. Durand, J. Dorsey, Fast bilateral filtering for the display of high-dynamic-range images, in: Proceedings of the 29th Annual Conference on Computer Graphics and Interactive Techniques, 2002, pp. 257–266.
[46] Z. Zhou, M. Dong, X. Xie, Z. Gao, Fusion of infrared and visible images for night-vision context enhancement, Appl. Opt. 55 (23) (2016) 6480–6490.
[47] A.S. Parihar, K. Singh, A study on Retinex based method for image enhancement, in: 2018 2nd International Conference on Inventive Systems and Control, ICISC, IEEE, 2018, pp. 619–624.
[48] C. Xydeas, V. Petrovic, Objective image fusion performance measure, Electron. Lett. 36 (4) (2000) 308–309.
[49] A.M. Eskicioglu, P.S. Fisher, Image quality measures and their performance, IEEE Trans. Commun. 43 (12) (1995) 2959–2965.
[50] J.W. Roberts, J.A. Van Aardt, F.B. Ahmed, Assessment of image fusion procedures using entropy, image quality, and multispectral classification, J. Appl. Remote Sens. 2 (1) (2008) 023522.
[51] V. Aslantas, E. Bendes, A new image quality metric for image fusion: The sum of the correlations of differences, Aeu-Int. J. Electron. Commun. 69 (12) (2015) 1890–1896.
[52] B. Shreyamsha Kumar, Multifocus and multispectral image fusion based on pixel significance using discrete cosine harmonic wavelet transform, Signal, Image Video Process. 7 (6) (2013) 1125–1143.
[53] H. Peng, F. Long, C. Ding, Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy, IEEE Trans. Pattern Anal. Mach. Intell. 27 (8) (2005) 1226–1238.
[54] Y.-J. Rao, In-fibre Bragg grating sensors, Meas. Sci. Technol. 8 (4) (1997) 355.
[55] Y. Han, Y. Cai, Y. Cao, X. Xu, A new image fusion performance metric based on visual information fidelity, Inf. Fusion 14 (2) (2013) 127–135.
[56] B. Shreyamsha Kumar, Image fusion based on pixel significance using cross bilateral filter, Signal, Image Video Process. 9 (5) (2015) 1193–1204.
[57] Y. Liu, X. Chen, R.K. Ward, Z.J. Wang, Image fusion with convolutional sparse representation, IEEE Signal Process. Lett. 23 (12) (2016) 1882–1886.
[58] W. Li, Y. Xie, H. Zhou, Y. Han, K. Zhan, Structure-aware image fusion, Optik 172 (2018) 1–11.
[59] Z. Zhou, B. Wang, S. Li, M. Dong, Perceptual fusion of infrared and visible images through a hybrid multi-scale decomposition with Gaussian and bilateral filters, Inf. Fusion 30 (2016) 15–26.
[60] H. Zhang, H. Xu, Y. Xiao, X. Guo, J. Ma, Rethinking the image fusion: A fast unified image fusion network based on proportional maintenance of gradient and intensity, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, (no. 07) 2020, pp. 12797–12804.
[61] M. Xu, L. Tang, H. Zhang, J. Ma, Infrared and visible image fusion via parallel scene and texture learning, Pattern Recognit. (2022) 108929.
[62] A. Toet, et al., TNO image fusion dataset, Figshare. data (2014).
[63] X. Jia, C. Zhu, M. Li, W. Tang, W. Zhou, LLVIP: A visible-infrared paired dataset for low-light vision, 2021, arXiv preprint arXiv:2108.10831.